Regression Assumptions Table
POLS 641 – F25

Violation	What it is - Assumption	Why a problem	Test Statistic	Visualization	Possible Solution
Omitted Variable Bias(Endogeneity)	IV uncorrelated with the error term	Biases our co-efficient and standard error	N/A (Look to theory and logic)	N/A (omitted variables is unmeasured)	Add the variable, instrumental variables, 2SLS, randomization
Non-Linearity (incorrect functional form)	there is a linear relationship between our IVs and DV.	Biases our co-efficient and standard errors	Likelihood ratio(LR) test, Ramsey reset	Scatter plots, residual plots to check residual vs fitted, compare lm to loess	Non-linear models(quadratic, cubic, etc), variable transformation
Outliers	No observations outside of the "universe of the population"	Can bias co-efficient, Impact r^2	Cooks Distance	Histogram, residual plots	Remove outlier if theoretically justifiable, Winsorizing, bad data entry?
Multicollinearity	the independent variables are not highly correlated with each other.	Faulty standard errors	Correlation Matrix, VIF (<4 no issue, >10 major issue)	Scatterplot of IVs	Take 1 variable out, find alternative measure(index?)
Autocorrelation (dependent error terms) in time-series	No correlation between the error terms	Faulty standard errors	Durbin-Watson (Between 2 and 4), Breusch Pagan	Residuals vs time plot, ACF plot	Add time to model, Robust standard errors, lag the DV, first difference,
Heteroskedasticity - (Inconstant variance)	For every value of the IV the error term has the same variance	Faulty standard errors	Breusch-Pagan, Shapiro-Wilk	Residual plots	Robust standard errors(White's, weighted least squares), log, additional variable, GLM model
Normality of Residuals	The residuals from each IV are normally distributed	Faulty Standard errors	Shapiro-Wilk	QQ Plot	Transformations, Robust standard errors

Other Research Design Issues to think about: measurement error, sample selection bias, model specification, reverse causality, etc.

*Please note this table is intended as a basic reference and is not comprehensive. It remains a work in progress.